Differences Between Amazon Kinesis Services

Amazon Kinesis Family

The Amazon Kinesis Family makes it easier to collect, process, and analyze real-time, streaming data so that you can get timely insights and react quickly to new information. Amazon Kinesis offers key capabilities to cost-effectively process streaming data at virtually any scale, along with the flexibility to choose the tools that best suit the requirements of your application. With Amazon Kinesis, you can ingest real-time data such as video, audio, application logs, website clickstreams, and Internet of Things (IoT) telemetry data for machine learning, analytics, and other applications. You can use Amazon Kinesis to process and analyze data as it arrives, which means that you can respond quickly instead of waiting until all your data is collected before it's processed.

For more information, see Amazon Kinesis.

Amazon Kinesis Data Streams

Amazon Kinesis Data Streams is a massively scalable and durable real-time data streaming service. Kinesis Data Streams is designed to continuously capture gigabytes of data per second from hundreds of thousands of sources, such as website clickstreams, database event streams, financial transactions, social media feeds, IT logs, and location-tracking events. The data that's collected is typically available in milliseconds, which means that you can use Kinesis Data Streams with use cases for real-time analytics, such as real-time dashboards, real-time anomaly detection, dynamic pricing, and more.

For more information, see Amazon Kinesis Data Streams.

Amazon Kinesis Data Firehose

Amazon Kinesis Data Firehose is designed to reliably load streaming data into data lakes, data stores, and analytics services. It can capture, transform, and deliver streaming data to Amazon Simple Storage Service (Amazon S3), Amazon Redshift, Amazon Elasticsearch Service, generic HTTP endpoints, and service providers like Datadog, New Relic, MongoDB, and Splunk. It is a fully managed service that automatically scales to match the throughput of your data and requires virtually no ongoing administration. It can also batch, compress, transform, and encrypt your data streams before loading, which minimizes the amount of storage that you use and increases security. Although it's easier to operate when compared with Amazon Kinesis Data Streams, Kinesis Data Firehose delivery streams have a higher latency from the moment that data is ingested. For example, you can set the batch interval to 60 seconds if you want to receive new data within 60 seconds of sending it to your delivery stream. However, you could have latencies that are lower than 1 second when you use Amazon Kinesis Data Streams.

For more information, see Amazon Kinesis Data Firehose.

Amazon Kinesis Data Analytics

Amazon Kinesis Data Analytics is designed to transform and analyze streaming data in real time with Apache Flink. Apache Flink is an open-source framework and engine for processing data streams. Amazon Kinesis Data Analytics reduces the complexity of building, managing, and integrating Apache Flink applications with other AWS services. Amazon Kinesis Data Analytics is designed to take care of everything that's required to run streaming applications continuously. It also scales automatically to match the volume and throughput of your incoming data. With Amazon Kinesis Data Analytics, there are no servers to manage, no minimum fee or setup cost, and you only pay for the resources that your streaming applications consume.

For more information, see Amazon Kinesis Data Analytics.

Amazon Kinesis Video Streams

Amazon Kinesis Video Streams is designed to securely stream video from connected devices to AWS for analytics, machine learning (ML), playback, and other forms of processing. Kinesis Video Streams automatically provisions and elastically scales the infrastructure that's needed to ingest streaming video data from millions of devices. It's designed to durably store, encrypt, and index video data in your streams, and you can access your data through the Kinesis Video Streams APIs. You can use Kinesis Video Streams to play back video for live and on-demand viewing. You can also use it to quickly build applications that take advantage of computer vision and video analytics through integration with Amazon Rekognition Video and libraries for ML frameworks (such as Apache MXNet, TensorFlow, and OpenCV). Kinesis Video Streams also supports WebRTC, an open-source project that uses APIs to facilitate real-time media streaming and interaction between web browsers, mobile applications, and connected devices. Typical uses for WebRTC include video chat and peer-to-peer media streaming.

For more information, see Amazon Kinesis Video Streams.

Mark as completed

Like Dislike Report an issue

coach

Hi, Venkatesh!
How can I help?

Explain this topic in simple terms
Give me a summary
Give me practice questions
Give me real-life examples

Ask me anything

Coach is powered by AI, so check for mistakes and don't share sensitive info. Your data will be used in accordance with Coursera's Privacy Notice.